

Smarter Access, Seamless Flow.

The Complete Ecosystem Architecture
for Entrance Automation.



The Architects of Automation

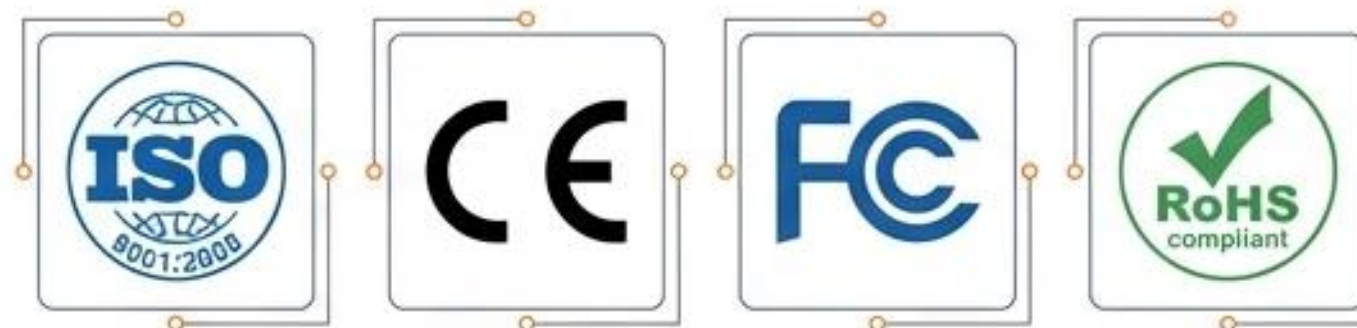
Legacy

20

Years of Industrial Excellence.

Proven operational experience across residential, commercial, and heavy industrial environments.

Quality Assurance



Government & Industry Trust

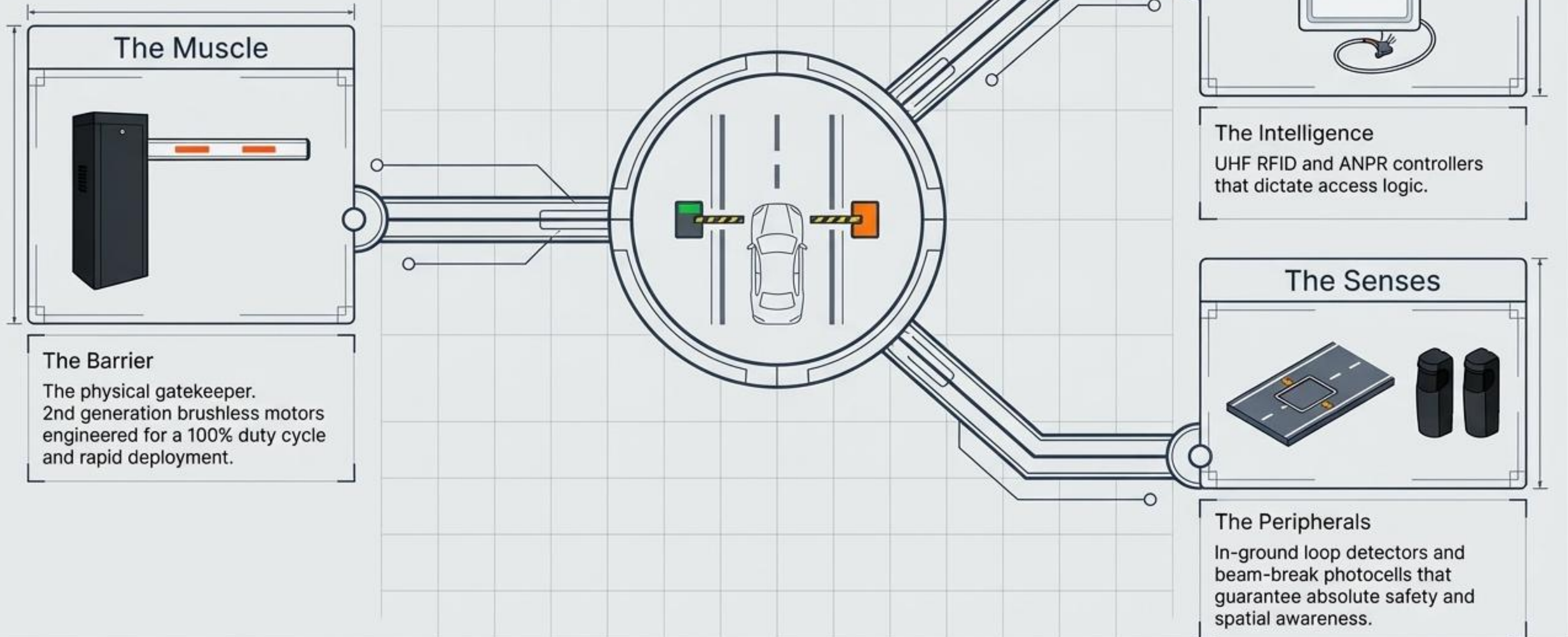


Mission

“ **Bridging manufacturing competency with technological agility to deliver high-security, 100% duty-cycle solutions.** ”



The SIE Ecosystem Architecture



The Core: 2nd Gen Brushless Boom Barrier



Endurance

100% Duty Cycle. Designed for relentless, non-stop operation in high-traffic zones. Backed by a 2-year manufacturing defect warranty.



Resilience

No-Clutch Design. Eliminates the most common mechanical failure point in traditional barriers.



Recovery

Instant Manual Override. Open the barrier via the motor wheel during power outages; system automatically resets upon power restoration.



Aesthetics

Customisable Chassis. Available in High-Visibility Orange, Caution Yellow, and Stealth Black.

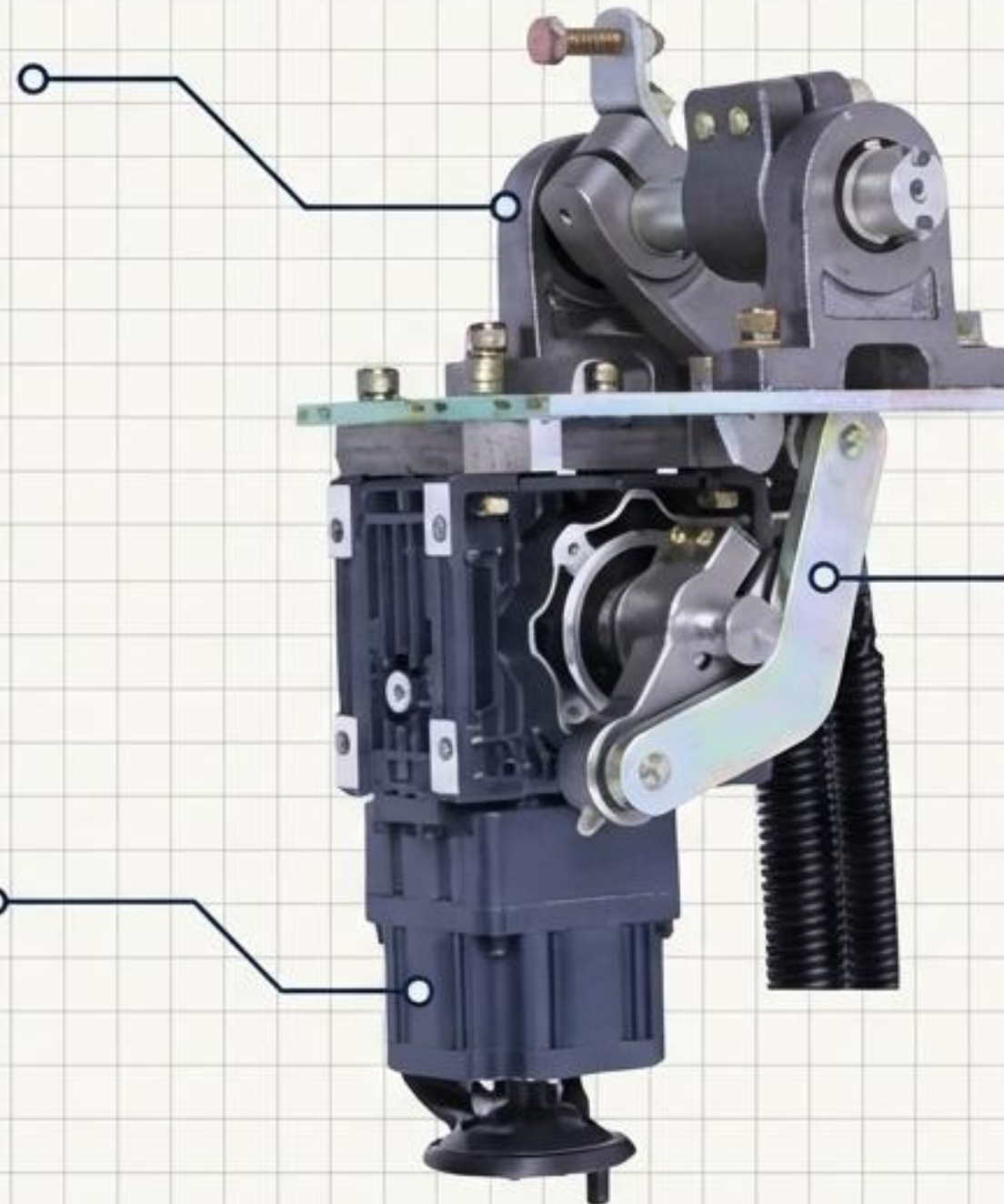
Mechanical Anatomy: Engineered for Zero Downtime

Curved Crank Arm Three-Link Structure

The Benefit: Guarantees fast, mathematically stable, and vibration-free landing of the boom rod.

Brushless Speed Control Motor

The Benefit: Frictionless operation allowing for precision speed calibration and silent running.



Two-Stage Worm Gear Transmission

The Benefit: Delivers massive torque (56.8 N.m) with minimal wear, significantly extending service life.

Performance Telemetry

1.5s - 8.0s

- **Inter:** Fully adjustable running speed for varying traffic flows.

140W

- Highly efficient DC24V consumption with an 8.58A rated current.

1850
RPM

- No-load speed generating 56.8 N.m of output power.

-35°C
to
+85°C

- **Endurance:** Extreme temperature tolerance for harsh Indian climates.

IP44

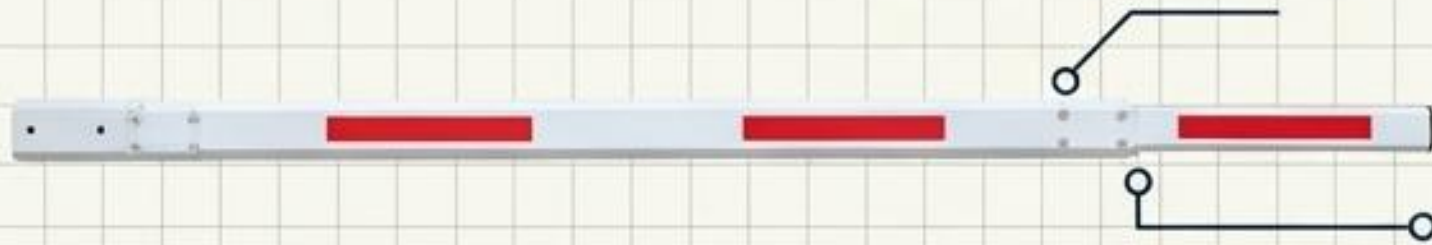
- **Protection:** Dust and water resistance grade. Humidity tolerance <90%.

100m

- **Control:** Standard wireless remote operation distance in open areas.

Spatial Flexibility: Boom Configurations

The Open Sky



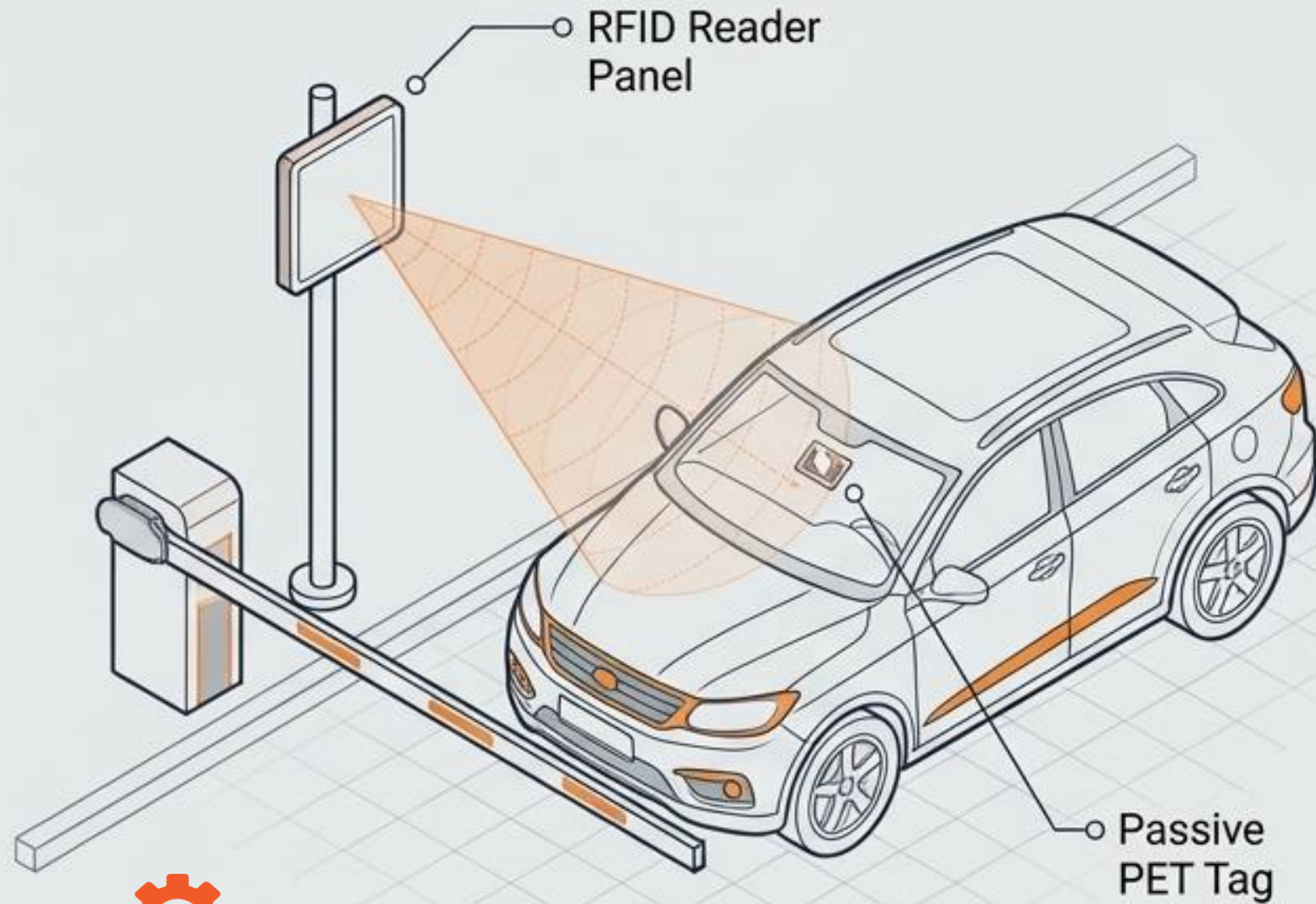
- **Mechanism:** Slide-to-extend design.
- **Variants:** 3.5m (extendable to 4.5m) and 4.5m (extendable to 6.2m).
- **Ideal For:** Toll plazas, main campus gates, and open-air commercial parking.

The Confined Space



- **Mechanism:** Central folding pivot with tension rod.
- **Variant:** Maximum horizontal reach of 4.5m.
- **Ideal For:** Underground residential basements and multi-level parking with low ceiling clearances.

The Brains: UHF RFID Intelligence



Frictionless Authentication

System Specs:

SIE88S & SIE88L Readers.
EPC Class 1 Gen2 protocol operating
on FHSS frequency hopping.
12dBi built-in polarised antenna.

Durability:

IP65 Rated. Fully ruggedised for outdoor
exposure (-25°C to +60°C).

The Handshake:

15-Metre Read Range. Passive PET windshield
tags (5g weight) create a frictionless *Answer Active
Mode* handshake, opening the gate before the
vehicle even stops. Up to 10m tag detection.

This diagram illustrates a vehicle barrier system with two detection methods. A white car is positioned behind a closed barrier. A red laser beam, labeled "Photocell Beam", originates from a sensor on the barrier and points towards the car. A "Loop Detector Grid" is embedded in the pavement in front of the barrier. Arrows indicate the direction of vehicle travel. Various dimensions are marked with arrows and labels: "A" for the barrier height, "B" for the distance from the barrier to the photocell sensor, "C" for the distance from the barrier to the loop detector grid, "D" for the width of the loop detector grid, and "E" for the distance from the photocell sensor to the loop detector grid. The car's dimensions are also indicated: "F" for the car's length, "G" for the car's width, and "H" for the car's height.

Available in 10m wired or wireless configurations.

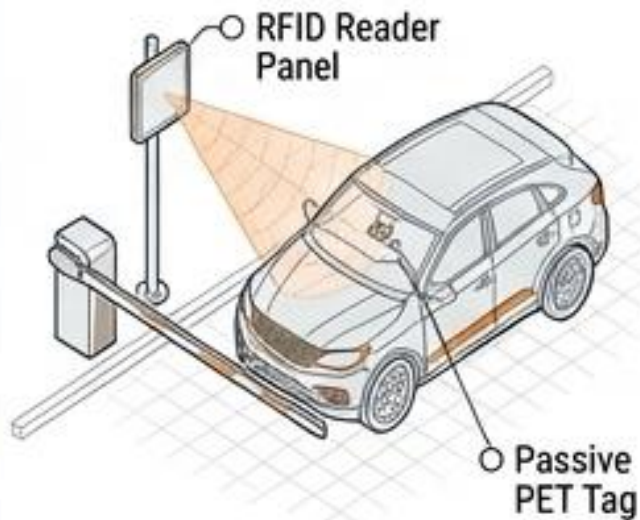
Features permanent presence holding (barrier will not close as long as metal is above the loop) and sensitivity reinforcement to avoid missing large trucks.

SYNTHESIS: The Cycle of Seamless Access

0.0s - Approach

The Brain Activates.

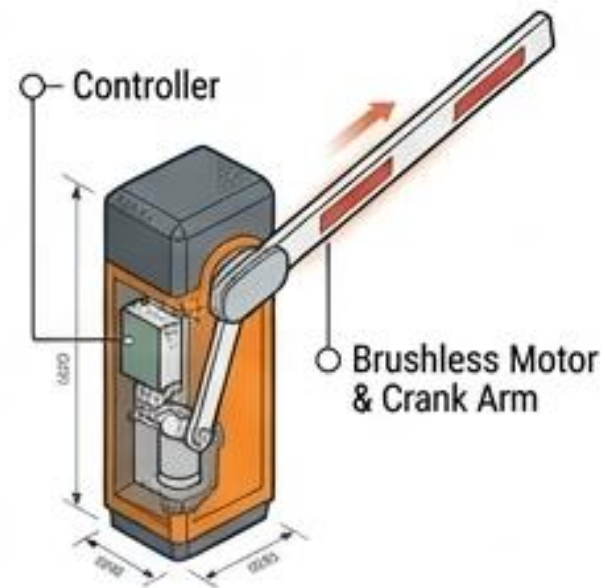
Vehicle enters the 15m radius.
UHF RFID Reader authenticates
the passive windshield tag.



0.5s - Action

The Muscle Engages

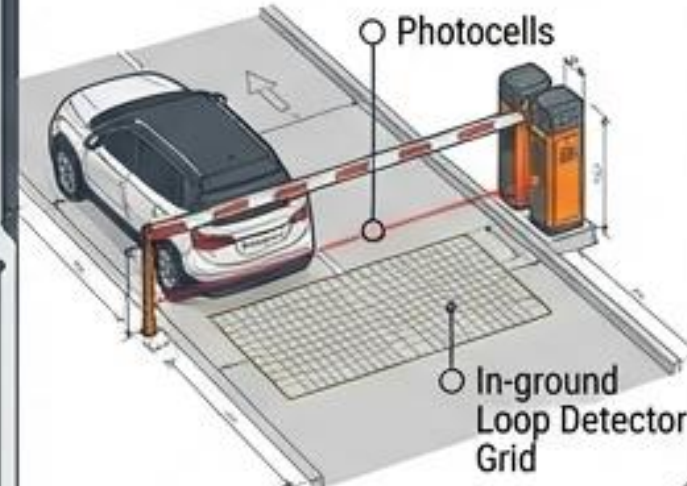
Controller sends dry contact signal.
Brushless motor powers the
3-link crank arm, lifting the
boom in 1.5 seconds.



2.0s - Safety

The Senses Monitor.

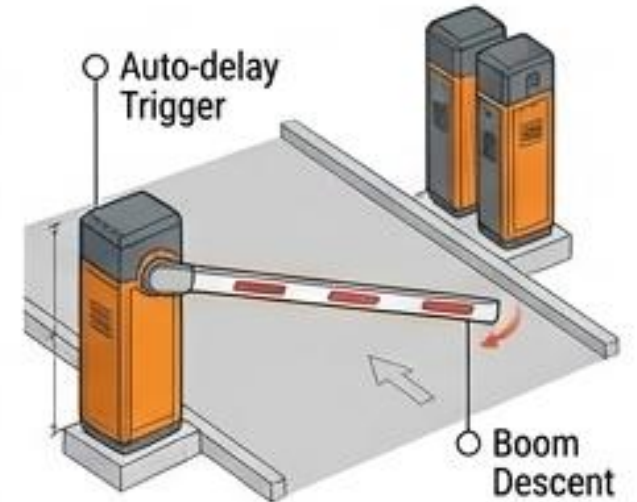
Vehicle passes through.
In-ground Loop Detector and
Photocells confirm physical
presence, overriding any close
commands to prevent crashes.



5.0s - Reset

The Cycle Completes

Vehicle clears the sensor footprint.
Auto-delay triggers the smooth
descent of the boom, securing
the perimeter.

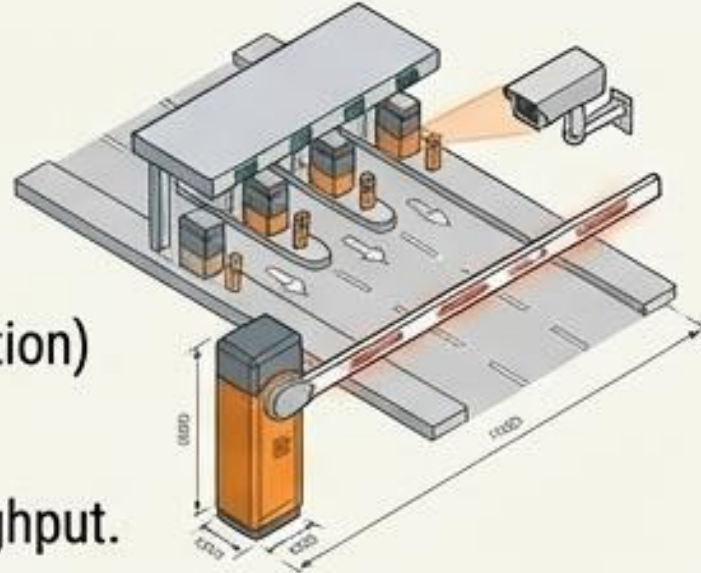


Application Architectures

Toll Management: The High-Speed Setup

3M Telescopic Boom + ANPR
(Automatic Number Plate Recognition)
Kit + Heavy-Duty Loop Detector.

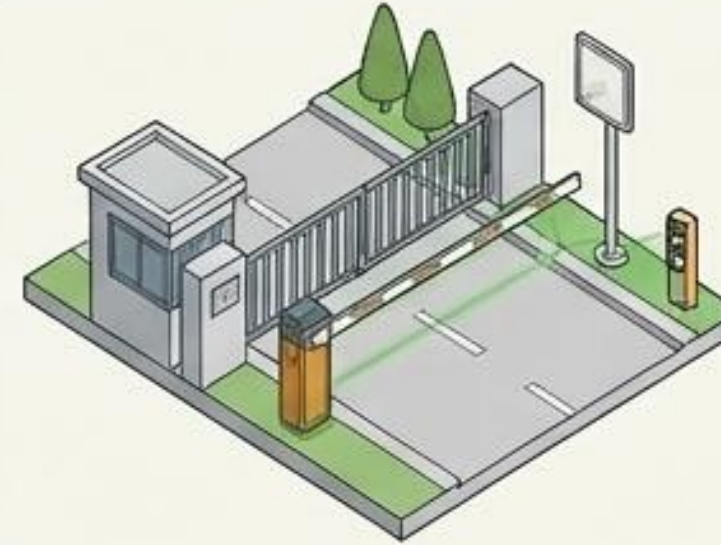
Optimised for sub-2-second throughput.



Residential Complex: The Frictionless Setup

4.5M Telescopic Boom +
15m UHF RFID + Wireless
Photocells.

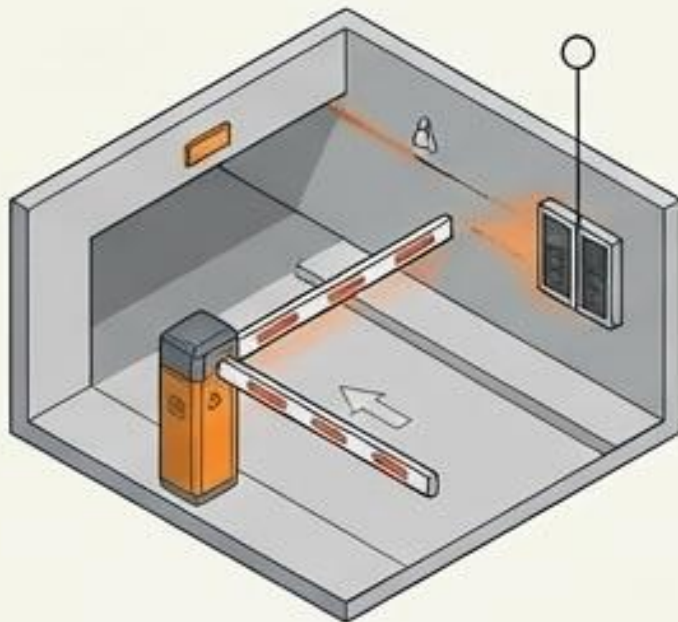
Optimised for zero-touch
resident entry.



Basement Parking Lot: The Confined Setup

4.5M Articulated 90-Degree
Boom + 2-Door Access
Controller.

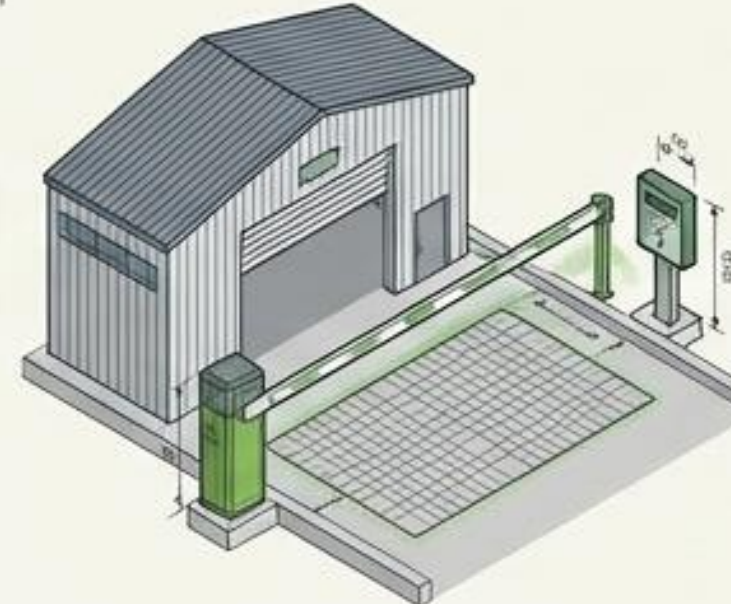
Optimised for low ceilings and
tight turning radiuses.



Industrial Warehouse: The Heavy-Duty Setup

6M Telescopic Boom + Key
Selector + Reinforced Loop
Senses.

Optimised for wide truck
turning circles and manual
oversight.





Secure Your Perimeter. Automate Your Flow.

Connect with our engineering and commercial teams to design a bespoke entrance architecture tailored to your spatial and traffic requirements.

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